

Power BI & DAX Cheat Sheet

A double-sided quick reference · DPM Learning Hub

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A printable reference for building models and writing DAX in Power BI. t = your table; 'Date' = a marked date table. Measures react to filters; columns don't.

POWER BI IN ONE PAGE	
Desktop	Build models, measures & reports (free).
Service (powerbi.com)	Publish, share, schedule refresh.
You'll make 3 things	Semantic model · reports · dashboards.
Build order	Get > Transform > Model > DAX > Visualize > Publish.

POWER QUERY (GET & SHAPE)	
Home > Transform data	Opens the query editor (same as Excel).
Change type / Rename	Click the type icon / double-click header.
Filter / Remove rows	Header dropdown · Remove Rows.
Split / Merge columns	Transform tab.
Unpivot	Wide > tall (tidy data for modelling).
Merge / Append	Join on a key / stack tables.
Rule	Shape in Power Query; calculate in DAX.

MODEL & RELATIONSHIPS	
Star schema	One fact table + dimension tables.
1-to-many	Dimension (one) > fact (many).
Single direction	Default filter flow; avoid both-ways.
Date table	Add one; Mark as date table.
Hide keys	Hide id columns from report view.

COLUMN VS MEASURE	
Calculated column	Per row, stored at refresh; group/filter by it.
Measure	Live, reacts to filters; all totals & KPIs.
Trap	A total/average that should react = measure.
Col = ... / Measure = ...	Both written in DAX, evaluated differently.

AGGREGATIONS	
SUM(t[col])	Total a column.
AVERAGE / MIN / MAX	Mean / smallest / largest.
COUNT(t[col])	Count non-blank values.
COUNTROWS(t)	Count rows in a table.
DISTINCTCOUNT(t[col])	Unique values.

CALCULATE — THE HEART	
CALCULATE(expr, filter1, ...)	Evaluate expr under modified filters.
CALCULATE([Rev], t[region]="EMEA")	Filter to EMEA.
CALCULATE([Rev], ALL(t))	Ignore filters (grand total).
Use FILTER for complex	CALCULATE([Rev], FILTER(t, t[x]>5)).

FILTER FUNCTIONS	
ALL(t) / ALL(t[col])	Remove filters from table/column.
ALLEXCEPT(t, t[k])	Remove all filters except the key.
REMOVEFILTERS()	Clear filters (modern ALL).
KEEPFILTERS(...)	Add a filter without replacing.
VALUES(t[col])	Distinct values in current context.

SAFE MATH & LOGIC	
DIVIDE(a, b, 0)	Safe division — no #DIV/0.
IF(test, yes, no)	Branch on a condition.
SWITCH(TRUE(), c1,r1, ..., def)	Clean multi-branch (vs nested IF).
a && b / a b	AND / OR.
COALESCE(a, b)	First non-blank value.
ISBLANK(x)	Test for blank.

ITERATORS (X FUNCTIONS)	
SUMX(t, expr)	Evaluate expr per row, then sum.
AVERAGEX / MAXX / MINX	Row-by-row then aggregate.
SUMX(t, t[qty]*t[price])	Row-level multiply then total.
RANKX(ALL(t[c]), [M])	Rank by a measure.

RELATIONSHIPS IN DAX	
RELATED(dim[col])	Pull a column from the one-side.
RELATEDTABLE(fact)	Related rows from the many-side.
USERELATIONSHIP(a, b)	Activate an inactive relationship in CALCULATE.

VAR / RETURN	
VAR x = SUM(t[rev])	Compute once, name it.
RETURN DIVIDE(x, ...)	Use the variable — clearer & faster.
Tip	Great for readable, debuggable measures.

TIME INTELLIGENCE	
Needs a Date table	Marked as date table.
TOTALYTD([Rev], 'Date'[Date])	Year-to-date.
SAMEPERIODLASTYEAR('Date'[Date])	Shift back one year.
DATEADD('Date'[Date], -1, MONTH)	Shift by a period.
PREVIOUSMONTH('Date'[Date])	Prior month.

COMMON MEASURES	
Total Rev = SUM(t[rev])	Base total.
Order Count = COUNTROWS(t)	Row count.
Delivered % = DIVIDE([Del], [Cnt])	A rate, safely.
YoY % = DIVIDE([Rev] - [RevLY], [RevLY])	Year-on-year growth.
Running = CALCULATE([Rev], DATESYTD(...))	Cumulative.

TEXT & FORMAT	
a & b / CONCATENATE(a,b)	Join text.
FORMAT(v, "0.0%")	Format a value as text.
LEFT/RIGHT/MID(s, n)	Slice text.
UPPER/LOWER(s) · LEN(s)	Case · length.

ROW VS FILTER CONTEXT	
Row context	'Current row' — in columns & iterators (X).
Filter context	Filters from slicers, rows, visuals.
CALCULATE	Turns row context into filter context.
Why blanks	A measure with no rows in context = BLANK.

CHOOSE THE VISUAL

Bar / column	Compare across categories (default).
Line	Trend over time.
Card / KPI	One headline number (+ target).
Matrix / table	Precise numbers, drill by row × col.
Scatter	Relationship between two measures.
Map	Values by geography.

REPORT UX

Slicers	Region / Date / Status filters for viewers.
Edit interactions	Control which visuals cross-filter.
Conditional formatting	Colour by value / rules.
Tooltips	Add detail on hover (even a mini page).
Bookmarks	Saved view states for navigation.

PERFORMANCE

Star schema	Fastest shape for the engine.
Measures, not columns	Less stored data, more flexible.
Avoid bi-directional	Use single-direction relationships.
Reduce cardinality	Fewer unique values = smaller model.
Hide unused columns	Trim the model; remove what you don't use.

GOTCHAS

Measure vs column	Totals/averages = measures.
Naked SUM in a card	Fine; but reuse via a named measure.
Filter context surprises	Wrap with CALCULATE to control it.
BLANK vs 0	Use COALESCE / +0 if you need a zero.
Implicit measures	Prefer explicit named measures.

PUBLISH & REFRESH

Publish	Desktop > Publish to a workspace.
Scheduled refresh	Set in the Service (dataset settings).
Gateway	Needed to refresh on-prem sources.
RLS	Row-level security filters by the viewer.
Apps	Package reports for an audience.

GET DATA (SOURCES)

Excel / CSV	File sources for quick models.
SQL / database	Import a table or a query.
Web / SharePoint	Pull from a URL or list.
Dataflows	Reusable cloud Power Query.
Import vs DirectQuery	Cached & fast vs live & large.

DAX TABLE FUNCTIONS

FILTER (t, cond)	Return a filtered table.
VALUES / DISTINCT (t[c])	Unique values as a table.
SUMMARIZE (t, t[c], ...)	Group into a summary table.
ADDCOLUMNS (tbl, "n", e)	Add computed columns.
TOPN (n, tbl, [m])	Top n rows by a measure.
CALENDAR (a, b)	Build a date table.

FORMAT A MEASURE

Measure tools > Format	Set %, currency, decimals.
FORMAT (v, "#,0")	Inline text format (returns text).
Data category	Tag URL / geo / image for visuals.
Display folders	Organise many measures.

HIERARCHIES & DRILL

Build a hierarchy	Group Year > Quarter > Month.
Drill down	Expand levels on a visual.
Sort by column	Sort Month by MonthNo, not text.
Groups / bins	Bucket values without DAX.

PRODUCTIVITY (DESKTOP)

Quick measures	Build common DAX from a dialog.
Performance analyzer	See what's slow on a page.
View > Themes	Apply consistent colours.
Templates (.pbit)	Reuse a model skeleton.